```
import matplotlib.pyplot as plt
import numpy as np
```

Question 5a

A complex sinusoid is denoted:

$$y(t) = e^{(\sigma + j\omega)t}$$

First compute a vector representing time from 0 to 10 seconds in about 500 steps (You can use np.linspace). Use this vector to compute a complex sinusoid with a period of 2 seconds, and a decay rate that reduces the signal level at 10 seconds to half its original value. What σ and ω did you choose? If your complex exponential is y, plot the signal.

We want $|y(10)| = \frac{1}{2}|y(0)|$. We have that

$$|y(t)| = |e^{\sigma t}||e^{j\omega t}| = |e^{\sigma t}|$$

Thus

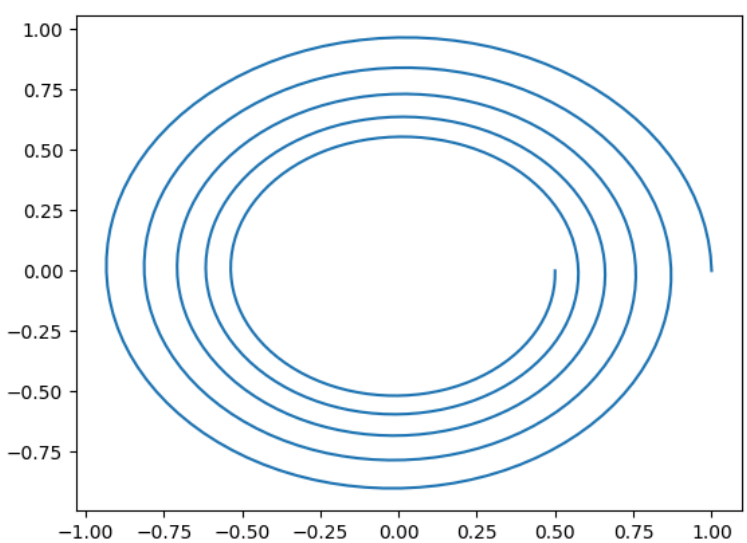
$$\begin{aligned} \frac{1}{2}|y(0)| &= \frac{1}{2} = e^{\sigma 10} \\ \sigma &= \frac{1}{10}\ln\left(\frac{1}{2}\right) \\ &= -\frac{\ln 2}{10} \end{aligned}$$

For ω, we want $T = 2$. Since $T = \frac{2\pi}{\omega}$,

$$\omega = \frac{2\pi}{2} = \pi$$

```
t = np.linspace(0, 10, 500)
sigma = -np.log(2) / 10
omega = np.pi
y = np.exp(sigma*t + 1j*omega*t)

# The code below will plot the signal on the real-imaginary complex plane
plt.plot(np.real(y), np.imag(y))
plt.show()
```

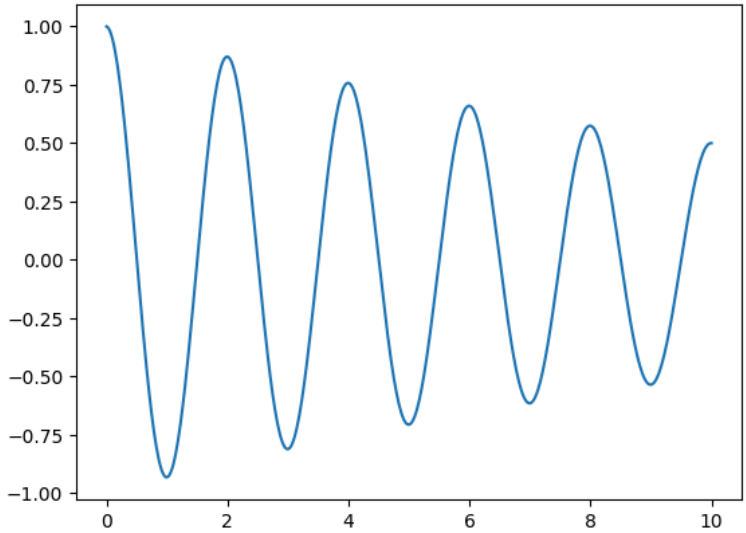


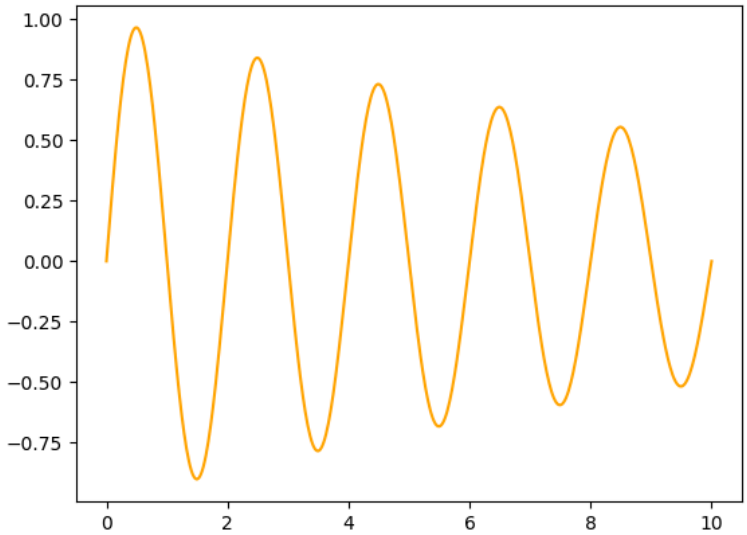
Question 5b

Use the np.real() and np.imag() functions to extract the real and imaginary parts of the complex exponential, and plot them as a function of time (plot them separately). This should look more reasonable. Label your axes, and check that your signal has the required period and decay rate.

```
real_part = np.real(y)
imag_part = np.imag(y)

# The code below will plot out the real and imaginary parts separately
plt.plot(t, real_part)
plt.show()
plt.plot(t, imag_part, color="orange")
plt.show()
```



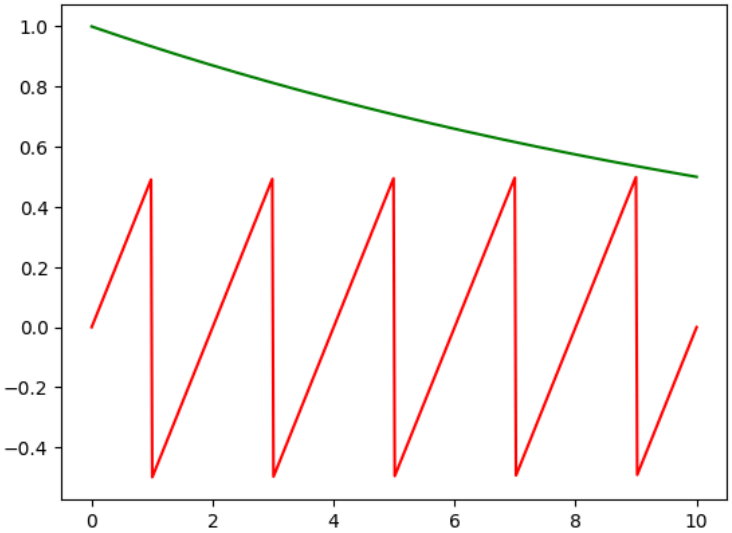


Question 5c

Use the np.abs() and np.angle() functions to plot the magnitude and phase angle of the complex exponential (plot them in the same figure). Scale the np.angle() plot by dividing it by 2π so that it fits well on the same plot as the np.abs() plot (i.e. plot the angle in cycles, instead of radians, the function np.angle(x) returns the angle in radians).

```
magnitude = np.abs(y)
angle = np.angle(y) / (2*np.pi)

# The code below will plot out the magnitude and phase parts on the same plot
plt.plot(t, magnitude, label='magnitude', color='green')
plt.plot(t, angle, label='phase', color='red')
plt.show()
```


1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
0
2
4
6
8
10